

Extended Simulation Results: Tables & Rules

1. Rules Used for Each Type (Base Classifier)

CST Dataset (F3 & F4)

- **Simulated:** IF F3 <= 0.00012364
- **Surface:** ELSE IF F3 <= 0.000562287 AND F4 <= 10.1456
- **Void:** ELSE IF F3 <= 0.000562287 AND F4 > 10.1456
- **Sharp Point:** ELSE IF F3 > 0.000562287 AND F4 <= 12.6887
- **Void (Fallback):** ELSE IF F3 > 0.000562287 AND F4 > 12.6887

BGST Dataset (F1 & F5)

- **Simulated:** IF F1 <= 7.89e-05
- **Surface:** ELSE IF F1 <= 0.0003135 AND F5 <= 197.5
- **Void:** ELSE IF F1 <= 0.0003135 AND F5 > 197.5
- **Surface (Fallback):** ELSE IF F1 > 0.0003135 AND F5 <= 65.4116
- **Sharp Point:** ELSE IF F1 > 0.0003135 AND F5 > 65.4116

HST Dataset (F1 & F5)

- **Simulated:** IF F1 <= 7.41e-05
 - **Surface:** ELSE IF F1 <= 0.0003325 AND F5 <= 200.5
 - **Void:** ELSE IF F1 <= 0.0003325 AND F5 > 200.5
 - **Sharp Point:** ELSE IF F1 > 0.0003325
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2. Classification Tables

Table 1: BGST Dataset Comparison

S.No	PD Types (BGST)	% Rule-Based (20dB)	% Rule-Based (50dB)	% SVM (20dB)	% SVM (50dB)	% CNN (20dB)	% CNN (50dB)
1	Simulated (116)	100.00	100.00	100.00	100.00	100.00	100.00
2	Sharp (116)	98.28	99.14	99.14	99.14	99.14	99.14
3	Void (253)	36.76	47.83	44.27	57.31	56.13	70.75
4	Surface (253)	98.02	97.63	97.23	89.33	84.58	77.47

Table 2: CST Dataset Comparison

S.No	PD Types (CST)	% Rule-Based (20dB)	% Rule-Based (50dB)	% SVM (20dB)	% SVM (50dB)	% CNN (20dB)	% CNN (50dB)
1	Simulated (116)	100.00	100.00	100.00	100.00	100.00	100.00
2	Sharp (116)	96.55	96.55	99.14	98.28	99.14	99.14
3	Void (253)	62.45	81.03	40.32	64.43	43.48	60.87
4	Surface (253)	66.01	45.06	81.82	61.26	80.24	65.61

Table 3: HST Dataset Comparison

S.No	PD Types (HST)	% Rule-Based (20dB)	% Rule-Based (50dB)	% SVM (20dB)	% SVM (50dB)	% CNN (20dB)	% CNN (50dB)
1	Simulated (116)	100.00	100.00	100.00	100.00	100.00	100.00
2	Sharp (116)	0.00	99.14	0.40	99.14	2.37	99.14
3	Void (253)	36.76	47.04	45.85	58.10	31.62	63.24
4	Surface (253)	98.81	98.81	96.84	90.12	94.86	84.19